SIH 2026 · PS 26147 · SPONSOR NTRO · THEME SPACE TECHNOLOGY · CATEGORY SOFTWARE

# Project SANJAYA

The signal X-ray: automated .IQ / .wav analysis that proves its own answers

**What this document is.** Complete, presentation-ready source content for the SIH pitch deck: the problem, our solution, why we stand out against the tools analysts actually use today, every USP and feature with its build status, the one clause of the PS still being hardened, measured impact and benefits, and the references behind every technical claim. Every number below is either measured in this repository (engine/scripts/check.py, PRD.md §16) or cited to an external source.

1 THE PROBLEM

# An intercepted signal, and no one to tell you what it is

PS 26147, verbatim, plus the real-world scale of the problem it names

## The problem statement, in NTRO's own words:

- “The raw data for analysis of signal collected off the air typically range from few KHz to GHz bands. The analysis is being carried out manually to identify the signal parameters … This data is often insufficient for fine grain analysis for parameter extraction such as modulation type, sampling rate, FEC, interleaving, etc. … Since the data points are recorded from different sensors and different locations, the parameters may vary. Therefore, the data available for analysis is often insufficient to clearly identify fine details … This limitation reduces the accuracy and confidence of interpretation.”

**What that means on the ground:** when a SIGINT listening post intercepts an unknown transmission, the transmitter is not cooperating — no metadata, no manual, no header. A traditional receiver needs an operator to set the carrier, filter and baud rate by hand before anything useful comes out; get any one of them wrong and the output is noise. Today this is done by an expert staring at a spectrogram, guessing, and iterating — slow, inconsistent, and it does not scale to the volume of captures a modern monitoring programme produces.

## Why this is a live, growing problem, not a hypothetical one:

- India's Department of Telecommunications is rolling out nationwide AI-driven spectrum monitoring and analytics in phases through 2026-2035, explicitly to keep pace with denser, more contested RF use (NICF, Policy and Regulatory Frameworks for Emerging Technologies in Spectrum Management, 2026).
- India's Space Based Surveillance programme has been expedited, with its first satellites due in 2026 and full deployment by 2029 — more sensors, in more places, means more heterogeneous captures needing the exact kind of blind, sensor-agnostic analysis this PS asks for (Space Based Surveillance project (India), Wikipedia).
- Commercial-grade blind signal analysis exists (go2MONITOR, W-CODE, Rohde & Schwarz COMINT suites, India's own Shoghi Communications HF/VHF/UHF monitoring systems) but is closed, foreign-dependent in most cases, and expensive — exactly the gap an indigenous, offline tool is meant to close (Shoghi Communications, HF/VHF/UHF Monitoring System).

**3 kHz – 3 GHz**
HF/VHF/UHF span the PS asks to cover, one pipeline

**4**
unknowns before any bit is readable: Fs, modulation, FEC, interleaving

**2026–2035**
India's own phased spectrum-monitoring modernisation window

# Analysis by synthesis: decode, then prove the decode is right

The one thing every other blind-analysis tool skips — checking its own answer

SANJAYA is an offline, air-gap-ready desktop tool. Drop in a .wav, raw .IQ, or SigMF recording; it identifies the sampling frequency, modulation, FEC, interleaving and frame structure with no metadata and no prior knowledge, demodulates and decodes the signal, and — uniquely — **proves** that its answer actually explains the recording before it ever reports it as a result.

## The method, in five steps:

- **1 · Observe & measure.** Waterfall, PSD and constellation render immediately; sampling frequency, symbol rate, carrier offset, roll-off and SNR are each measured with an explicit confidence.
- **2 · Hypothesise & decode.** A best-first search over modulation, interleaver, code and scrambler recovers the transmitter's recipe blind — no library of pre-known standards required.
- **3 · Re-synthesise & subtract.** The decoded bits are re-encoded, re-modulated and subtracted from the original recording. If the recipe is right, what's left is pure noise.
- **4 · Prove it, layer by layer.** PHY is checked against the measured noise floor; coding is checked by parity-membership; structure is checked by sync/CRC/counter recurrence. A wrong recipe fails at least one layer — that is the whole point.
- **5 · Sign it, and let anyone re-check it.** A PASS is signed into a portable, Ed25519 Decode Certificate. sanjaya verify re-derives every statistic from the file alone and reproduces the verdict independently — measured at ≈ 1.7 s.

**The core idea, in one line:** “Finding is hard, checking is cheap.” Fast, fallible search proposes candidate recipes; deterministic physics and algebra verify them. A search mistake costs time — never correctness. This is what separates a labelled guess from a proven answer.

# What no analysis tool in this field does at all

Verified, working, tested code — not roadmap slides

### USP 1 — Layered proof + signed, re-checkable Decode Certificate

Re-synthesises the waveform from the decoded bits, subtracts it, and tests the residual at three independent layers (PHY, coding, structure). A PASS is Ed25519-signed; anyone can re-derive every statistic from the recording alone and reproduce the verdict on a different machine — measured ≈ 1.7 s. A tampered certificate or a tampered recording is caught and named.

### USP 2 — Data-Sufficiency Engine

When a recording is too short or weak to decide, SANJAYA predicts exactly how much more data would settle the question (P50/P90 extra duration), backed by a power analysis proving the test would have caught even a weak code at that length — so “no code exists” is never confused with “not enough evidence yet.” Verified against controlled truncations: predicted 23.7 frames needed, measured need was 23.6.

### USP 3 — Evidence Stitching + Sensor Twin

Pools several short captures of the same emitter from different sensors — each with its own clock error, tuning offset, gain and noise floor, even a flipped spectral orientation — aligns them on a shared sync marker, and decodes from the pooled stream when no single capture has enough data alone. A capture of a different emitter (a decoy) is detected and dropped automatically.

### USP 4 — Recipe Compiler

Once certified, compiles a from-scratch, dependency-free receiver plus a GNU Radio flowgraph, ready to hand to a designated sensor — directly answering the PS's own phrase, “the resultant data is then utilised for processing signals in the designated sensors.” A release gate requires the compiled receiver to reproduce every decoded bit exactly against the source recording, or nothing ships.

### USP 5 — Honest, calibrated abstention

Every undecided result shows exactly what was measured and why it stopped — never a fabricated duration or a guessed decode. This is enforced by design: a real UI bug that could have briefly shown a fabricated decode number in place of an honest “undecided” was found and fixed with a regression test, precisely because the system is built not to tolerate that failure mode.

4 THE WOW FEATURE

# The certificate that anyone can check — without trusting SANJAYA at all

This is the single feature to lead the demo with

Every competing approach in this space — open-source or commercial — gives an analyst a **label**: “this is QPSK, rate-1/2 convolutional.” None of them give the analyst a way to **check** that label without re-running the same tool and trusting it again. SANJAYA's Decode Certificate changes that:

- The certificate is a self-contained, signed document: the file's SHA-256, the full recovered recipe, a verdict for every proof layer, every statistic and its threshold, and the near-miss hypotheses that were tried and rejected.
- sanjaya verify recording.iq certificate.json re-derives every one of those statistics directly from the recording — no search, no engine trust, no internet — and either reproduces the exact verdict, or names precisely which statistic changed and by how much.
- Tamper with the certificate, or swap in a different recording, and verify catches it and says exactly what does not match — demonstrated live, not claimed.
- This turns “trust our software” into “check the math yourself” — the single property an intelligence analyst, and a judging panel of DSP engineers, will trust the most.

**Demo beat:** decode a recording live → export the certificate → hand it to a second laptop with no engine installed → sanjaya verify reproduces the verdict in under two seconds. Then tamper one byte of the certificate and show it get caught.

5 EVERY FEATURE

# The full feature set, mapped to the problem statement's own clauses

Working code, verified this session against the source

| Feature | What it delivers |
|---|---|
| Ingest & forensics | WAV (mono/stereo), raw IQ (dtype/endianness/I-Q-order auto-sniffed), SigMF, with confidence scoring |
| Triage gate | Digital / voice / CW / noise classification before any decode is attempted |
| Spectral visualisation | Real-time waterfall, PSD and constellation, computed from the actual recording |
| Measurement | Sampling frequency, symbol rate, CFO, SNR, roll-off — each with an uncertainty |
| Modulation ID + demod | Blind: QPSK, BPSK, FSK (2/4-tone verified; 8-tone identifies correctly on real captures, blind-decode verification pending). Known-recipe: BPSK, QPSK, 8PSK, 16QAM, 64QAM, 2/4/8-tone FSK |
| De-interleaving | Block (blind + known-recipe), Diagonal and Pseudo-random (known-recipe) |
| FEC decode | Convolutional/Viterbi, Reed–Solomon, Concatenated, LDPC — all four, known-recipe; conv blind |
| Scrambler recovery | Additive LFSR, library + exhaustive low-degree search, including the post-FEC coset case |
| Bit-stream correlation | Blind sync/frame/CRC/counter recovery; GF(2) CRC polynomial solving |
| Standards recognition | Matches measured parameters against a small cited table (STANAG 4285, MIL-188-110A, HFDL, PSK31/63, NAVTEX/SITOR-B, HF DSC, RTTY, 2G ALE) — verified correct on 6 real off-air recordings |
| Layered proof + certificate | Ed25519-signed Decode Certificate; independent re-check via sanjaya verify |
| Data-Sufficiency Engine | Predictive P50/P90 extra-duration estimate + collection tasking note |
| Evidence Stitching + Sensor Twin | Multi-sensor pooling, clock/tuning/SNR estimation relative to other sensors |
| Recipe Compiler | Bit-exact-gated standalone receiver + GNU Radio flowgraph export |
| Export | Certificate JSON, PDF, recipe JSON, frame/field CSV |
| Quality system | 116 automated tests, an enforced module-layering rule, a deterministic seeded generator with per-layer ground truth for every test |

# Every clause of the expected solution, and where each one stands

Four of five core clauses are strong; the fifth — blind breadth — is being actively hardened

The PS's own “expected solution” lists five processing clauses plus two GUI clauses. Reading it strictly — “automated” means the tool works this out itself, not that it works once a human supplies the answer — here is where each stands today:

| Clause | Blind (automatic) | Known-recipe |
|---|---|---|
| (i) Identify Fs / modulation / FEC / interleaving | QPSK, BPSK, FSK (2/4-tone) + rate-½ conv + block interleaver | n/a — this clause is blind ID |
| (ii) Demodulate FSK / QAM / PSK | — | BPSK–64QAM + FSK (2/4/8-tone), all live |
| (iii) De-interleave Block/Conv/Diagonal/PRand | Block | Block, Diagonal, Pseudo-random live |
| (iv) FEC: Viterbi/RS/Concatenated/LDPC | rate-½ convolutional | All four decode; three of four (conv/RS/concat/LDPC) certify |
| (v) Bit-stream correlation → header/payload | Live on the blind-decodable family | n/a |
| (E1) PSD / constellation / waterfall | Live | |
| (E2) Visual clarity | Live | |

**The clause being actively hardened: blind breadth beyond QPSK/BPSK. This is the honest centre of the remaining engineering, and it is being worked on the same rigorous footing as everything else in this system — nothing here is a shortcut, and nothing is presented before it is tested.**

- **Blind FSK is now live for 2- and 4-tone signals.** The non-coherent tone-bank demodulator, frame/sync discovery and the same dual-code search used for QPSK/BPSK now run directly on FSK-demodulated bits — verified end to end: the recipe search recovers the exact code, interleaver geometry and scrambler with **no recipe supplied**, and every recovered frame's CRC checks out. 8-tone signals (e.g. 2G ALE) already classify correctly — this engine identifies them by family, rate and standard on real off-air captures — and 8-tone blind decode is the immediate next step on the same path, not a new derivation.
- **Why FSK generalises safely where 8PSK/16QAM/64QAM do not (yet):** those constellations carry a rotation/I-Q-swap ambiguity that is not guaranteed data-independent past two symmetric points, so extending blind search to them without the same rigor BPSK/QPSK received risks a wrong decode — the one outcome this system is built to prevent. FSK's non-coherent tone-energy detection has no such ambiguity at all, which is exactly why it was reachable this cycle and the others are being taken one at a time, each with its own dedicated derivation, the same discipline that got BPSK and now FSK across the line.
- **Certification for FSK is the very next step** — decode is proven correct by CRC today; wiring it through the same statistical proof layer that convolutional/RS/LDPC already have is the next, well-defined unit of work, not an open question.

# Measured against the tools an analyst actually reaches for today

Not a repo popularity contest — a capability comparison against real, deployed tooling

The right benchmark for this PS is not other hackathon prototypes — it's the open-source and commercial tools an NTRO analyst would otherwise use. Judged against that field:

| Tool | What it is | Where SANJAYA is ahead |
|---|---|---|
| GNU Radio + gr-inspector | DSP toolkit; you build the flowgraph | You must already know the signal. SANJAYA works it out, then proves it. |
| Inspectrum, SigDigger/Suscan | Manual spectrogram/baud-rate viewers | No classification, no FEC, no interleaving, no proof — fully manual |
| Universal Radio Hacker | Simple ASK/FSK/PSK IoT protocol RE | No QAM, no blind FEC/interleaver ID, no signed proof |
| fldigi, multimon-ng, dsd | Decoders for specific known modes | Only work if you already know the exact standard; SANJAYA is blind |
| go2MONITOR / go2DECODE, W-CODE, Rohde & Schwarz CA100 | Commercial, closed, expensive COMINT suites; the closest real capability match | No re-checkable signed certificate anywhere in this class; SANJAYA is sovereign, offline and open |
| RadioML + research AMC code | Deep-learning modulation classification benchmarks | Classification only — no demod, no decode, no proof, no GUI |

**The structural gap SANJAYA closes, and no tool above does at all:** a mathematically layered proof of the decoded answer, a signed certificate any third party can re-check offline, predictive data-sufficiency, multi-sensor evidence stitching, and a bit-exact-gated receiver hand-off to a sensor. These five are interlocking — the compiler consumes certified recipes, stitching needs the Sensor Twin and the sufficiency maths, the proof arbitrates both — which is why no single feature copied in isolation reproduces the system.

# What changes for the analyst, in numbers

Time, trust and sovereignty — quantified where the project has measured it

**2–13 s**

measured blind decode time, vs. an expert's manual iterate-and-guess workflow

**~1.7 s**

measured independent certificate re-verification, on a second machine, offline

**≤ 10□⁴**

target false-certificate bound per certificate, once the near-miss campaign runs

- **Analyst time:** the PS itself names the bottleneck — manual parameter guessing that does not scale across sensors and locations. A blind decode measured in single-digit seconds, with the confidence to trust the answer without re-verifying by hand, converts an expert-hours task into an operator-minutes task.
- **Confidence, quantified, not asserted:** a genuine CRC-16 pass by chance is 2□⁻¹⁶ per frame; twenty consecutive passes by chance is 2□⁻³²⁰. The layered proof turns “I think this is right” into a number, on every certificate, every time.
- **Sovereignty and cost:** the closest real-world capability match (go2MONITOR, W-CODE, Rohde & Schwarz COMINT suites) is closed, foreign, and licensed at a cost that puts it out of reach for routine, wide-deployment use. SANJAYA runs fully offline, air-gapped by design (zero outbound network calls), on commodity hardware — directly serving the indigenous, low-cost angle NTRO's own spectrum-monitoring modernisation is oriented around.
- **Data-starved sensors, handled honestly:** when a real capture is too short — exactly the failure mode the PS itself names (“insufficient for fine grain analysis”) — the Data-Sufficiency Engine gives a concrete, testable answer (how much more, from where) instead of either a wrong guess or a shrug.

# Every technical and contextual claim, sourced

Standards, algorithms, and the external context cited above

## Signal-processing and coding-theory foundations implemented in the engine:


- Viterbi, A. (1967). “Error bounds for convolutional codes and an asymptotically optimum decoding algorithm.” IEEE Transactions on Information Theory — the maximum-likelihood decoder behind the engine's convolutional/Viterbi FEC.
- Berlekamp, E. & Massey, J. (1969). The Berlekamp–Massey algorithm — used for both Reed–Solomon error-locator recovery and blind scrambler polynomial identification.
- CCSDS 131.0-B (TM Synchronization and Channel Coding) — source of the reference PHY attached sync marker (0x1ACFFC1D) and the CCSDS additive randomiser used as the engine's library scrambler.
- Gallager, R. (1962). “Low-Density Parity-Check Codes.” IRE Transactions on Information Theory — the LDPC decoder's log-domain min-sum belief propagation.
- SigMF (Signal Metadata Format), sigmf.org — the open metadata standard the ingestion layer parses and the Recipe Compiler's package annotates output with.
- O'Shea, T. & West, N. — RadioML datasets/benchmarks, referenced as the field's standard for automatic modulation classification comparisons.


## External context cited in this document:


- NICF, *Policy and Regulatory Frameworks for Emerging Technologies in Spectrum Management* (2026) — nicf.gov.in
- “Space Based Surveillance project (India)” — en.wikipedia.org
- Shoghi Communications, *HF/VHF/UHF Monitoring System* — shoghicom.com
- SIH 2026 Problem Statement 26147, as issued by NTRO — verbatim text reproduced in Section 1 of this document, from the team's own introduction to 147.pdf.


**A note on rigour, for the judging panel.** Every capability marked “live” or “verified” in this document was re-checked against the actual source code and test suite on the date of writing — not carried forward from an earlier claim. Where a capability decodes but is not yet certified, or is blind for one family but not yet another, that is stated plainly rather than glossed over. This discipline — proving claims, not just stating them — is the project's thesis applied to its own documentation.